Ontology-Guided Explainable Deep Learning for Arsenicosis Detection

Beyond Binary Classification through Clinical Reasoning

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Introduction

Chronic arsenic exposure through contaminated groundwater affects 20-40 million people in Bangladesh, first appearing as skin lesions before progressing toward cancer. Existing CNN screening tools output only a binary affected/healthy label, with no clinically interpretable reasoning behind the prediction.

Methods

Our pipeline links three components to move beyond a binary label toward an explainable, clinically-grounded output:

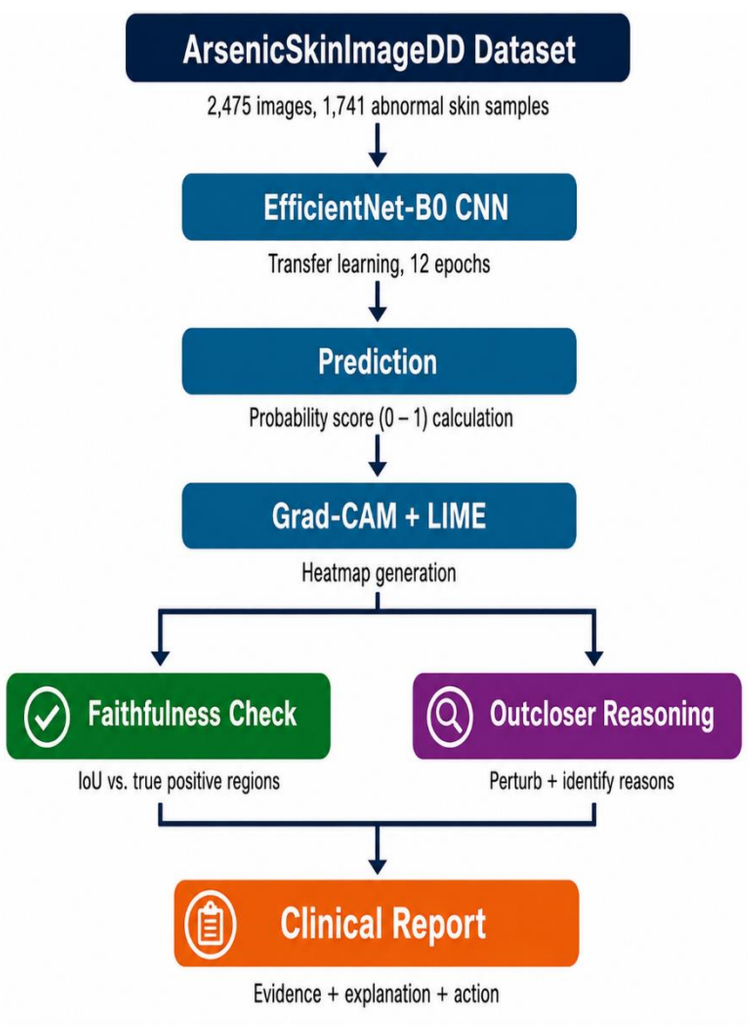
CNN CLASSIFICATION & EXPLAINABILITY

- EfficientNetB0 (frozen backbone) fine-tuned on 1,287 original images (85/15 train/validation split).
- Grad-CAM and LIME identify what lesion pattern is present and where it is located

Table I — Dataset Summary

Attribute	Value
Dataset	ArsenicSkinImageBD (Emu et al., 2023)
Classes	2 (Arsenic-Affected, Healthy)
Original images (spec)	1,482 (741 + 741)
Images used (available)	1,287 (741 affected, 546 healthy)
Source	4 field sites, Chapainawabganj, Bangladesh
Train / Validation split	85% / 15% (1,093 / 194 images)

System Workflow (Methodology)



ONTOLOGY-GUIDED REASONING

- OWL ontology built in Protege, verified via HermiT reasoner.
- Literature-grounded rules link lesion type + location to clinical risk.

Data Analysis

We quantify whether Grad-CAM attention overlaps the true lesion location using an IoU-based faithfulness score (Fig. B). Steps:

A. Manually box-annotate lesion regions in 18 sample images.

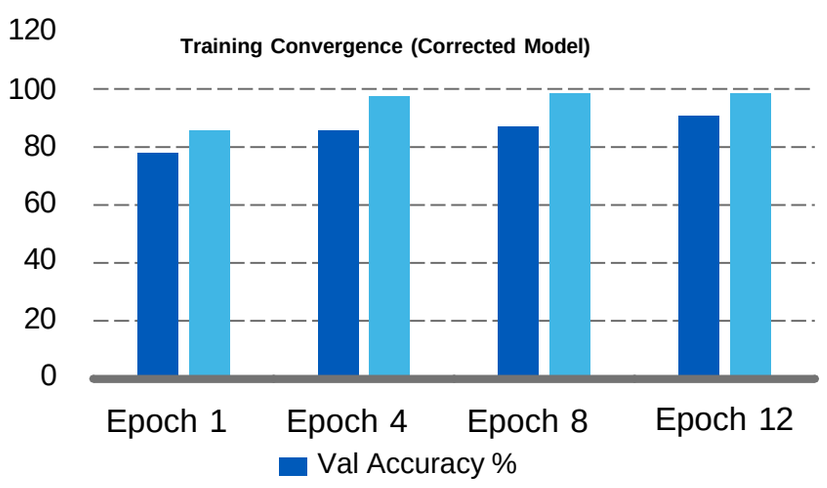
B. Compute IoU between Grad-CAM high-activation region and box.

C. Result: mean IoU = 0.144 (median 0.070, range 0.018-0.502).

Grad-CAM Explanation Quality (IoU)

Image	Confidence	Faithfulness	Note
IMG_1184	81.5%	0.502	Best overlap
IMG_1176	97.9%	0.339	Partial overlap
IMG_1173	99.1%	0.322	Diffuse pattern
Mean (n=18)	94.6%	0.144	Overall score

**IoU between Grad-CAM high-activation region and manually annotated lesion box.*



Results

Our final EfficientNetB0 model reaches 90.7% validation accuracy and 0.984 validation AUC after 12 epochs (85/15 split, corrected labels), with train/validation metrics tracking closely (Table 1) — no sign of overfitting despite the small dataset.

Grad-CAM/LIME localize lesions correctly in high-confidence cases; the faithfulness score confirms this quantitatively and flags cases where attention drifts to image corners instead of the visible lesion.

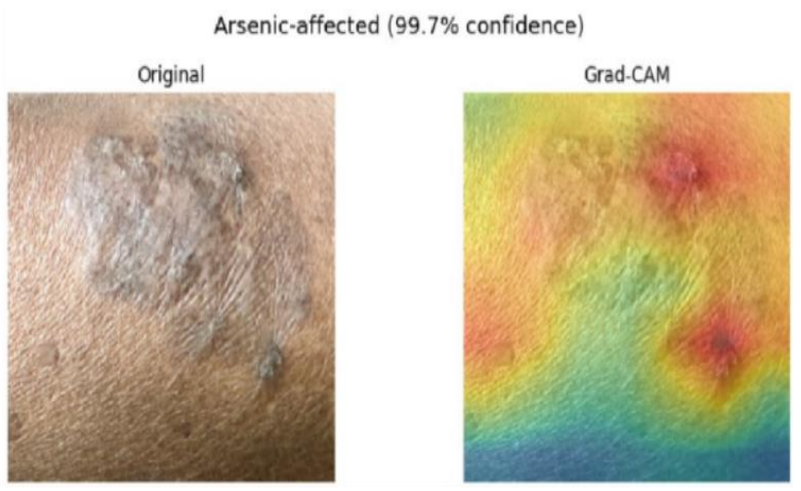


Figure A. Grad-CAM heatmap over a keratotic lesion (IMG_1516, 99.7% confidence).

Ontology Verified via HermiT

DL Query 'Trunk Keratosis' automatically returns real detected lesions (IMG_1176, IMG_1184) via HermiT reasoning, confirming the pipeline works end-to-end.

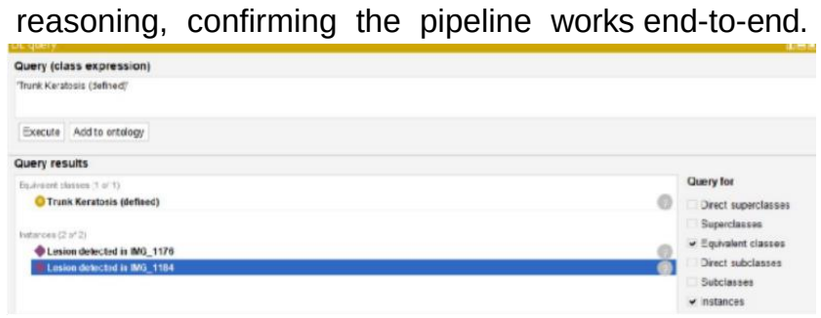


Figure B. Protege DL Query 'Trunk Keratosis' auto-classifies real detected lesions after HermiT reasoning.

Conclusion

We built a working CNN + Grad-CAM + Ontology pipeline that turns a binary label into a clinically-grounded explanation, validated on real dataset images:

- **Working End-to-End Pipeline:** CNN, Grad-CAM/LIME, and a Protege ontology verified together on real images.
- **Key Limitation:** lesion type/location is currently identified manually from Grad-CAM output; automating this is future work.

References

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2. Guha Mazumder, Diagnosis of Arsenicosis, PubMed 12635831
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4. J Pak Assoc Dermatol 2014;24(4):286-291
5. PathologyOutlines.com, Arsenical Keratosis
6. Emu et al., ArsenicSkinImageBD, Data in Brief, 2023
7. Mehedi et al., ArsenicNet, 2025 (97.69% binary accuracy)

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Table II — Training Hyperparameters

Hyperparameter	Value
Input resolution	224 × 224 × 3
Backbone	EfficientNetB0 (frozen, ImageNet)
Batch size	32
Optimizer	Adam
Learning rate	1e-4
Loss function	Binary Cross-Entropy
Dropout	0.4
Epochs	12 (early stopping)

Table III — Classification Report

Metric	Value
Accuracy	90.7%
AUC	0.984
Precision	0.918
Recall	0.878
F1-score	0.935

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